Name

Mathematics Advanced Year 12

Using Derivatives

Book 4

Version: 261021

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Syllabus Content

MAV-12-04 selects and applies differentiation methods to solve problems

Using derivatives

- Apply the product, quotient and chain rules to differentiate functions of the form $f(x)g(x)$, $\frac{f(x)}{g(x)}$ or $f(g(x))$, where $f(x)$ and $g(x)$ are any of the functions within the scope of the Mathematics Advanced course
- Solve problems involving equations of tangents and normals to curves involving any of the functions within the scope of the Mathematics Advanced course

Mixed Differentiation Practice

ⓘ Derivative Formulas

Function	Derivative
$y = f(x)^n$	$\frac{dy}{dx} = n\,f'(x)\,[f(x)]^{n-1}$
$y = f(x) + g(x)$	$\frac{dy}{dx} = f'(x) + g'(x)$
$y = k \cdot f(x)$	$\frac{dy}{dx} = k \cdot f'(x)$
$y = g\big(u(x)\big)$	$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$
$y = uv$	$\frac{dy}{dx} = u'v + uv'$
$y = \frac{u}{v}$	$\frac{dy}{dx} = \frac{u'v - uv'}{v^2}$
$y = \sin f(x)$	$\frac{dy}{dx} = f'(x)\cos f(x)$
$y = \cos f(x)$	$\frac{dy}{dx} = -f'(x)\sin f(x)$
$y = \tan f(x)$	$\frac{dy}{dx} = f'(x)\sec^2 x$
$y = e^{f(x)}$	$\frac{dy}{dx} = e^{f(x)}$
$y = \ln f(x)$	$\frac{dy}{dx} = \frac{f'(x)}{f(x)}$
$y = a^{f(x)}$	$\frac{dy}{dx} = f'(x)(\ln a)a^{f(x)}$
$y = \log_a f(x)$	$\frac{dy}{dx} = \frac{f'(x)}{(\ln a)f(x)}$

1 Differentiate with respect to x.

a e^{7x} — $7e^{7x}$

b e^{-x} — $-e^{-x}$

c e^{6x-2} — $6e^{6x-2}$

d $x + \ln x$ — $1 + \frac{1}{x}$

e $1 - \ln 3x$ — $-\frac{1}{x}$

f $\ln(3x+1)$ — $\frac{3}{3x+1}$

g $\sin 4x$ — $4\cos 4x$

h $\cos 3x$ — $-3\sin 3x$

i $\tan 5x$ — $5\sec^2 5x$

j $\ln(x^2-4)$ — $\frac{2x}{x^2-4}$

k $\ln(5x^3+3x-9)$ — $\frac{15x^2+3}{5x^3+3x-9}$

l $\ln(5x+1) + x^2$ — $\frac{10x^2+2x+5}{5x+1}$

m e^{-2x} — $-2e^{-2x}$

n e^{10x} — $10e^{10x}$

o $e^{2x} + x$ — $2e^{2x} + 1$

p $\tan(3x+1)$ — $3\sec^2(3x+1)$

q $\cos(-x)$ — $\sin(-x)$

r $3\sin x$ — $3\cos x$

s e^{x^2+1} — $2xe^{x^2+1}$

t e^{x^3+5x+7} — $(3x^2+5)e^{x^3+5x+7}$

u e^{5x} — $5e^{5x}$

v $(\ln x)^4$ — $\frac{4}{x}(\ln x)^3$

w $(x^2+\ln x)^6$ — $6\left(2x+\frac{1}{x}\right)(x^2+\ln x)^5$

x $x\ln x$ — $1 + \ln x$

y $x^2 + 2x + e^{1-x}$ — $2x + 2 - e^{1-x}$

z $(x+e^{4x})^5$ — $5(1+4e^{4x})(x+e^{4x})^4$

aa xe^{2x} — $e^{2x}(2x+1)$

bb $3x^2 + 5x - 5 + \ln 4x$ — $6x + 5 + \frac{1}{x}$

cc $\ln(8x-9) + 2$ — $\frac{8}{8x-9}$

dd $\ln(2x+4)(3x-1)$ — $\frac{6x+5}{(x+2)(3x-1)}$

ee $4\cos(5x-3)$

$-20\sin(5x-3)$

ff $2\cos(x^3)$

$-6x^2\sin(x^3)$

gg $7\tan(x^2+5)$

$14x\sec^2(x^2+5)$

hh $\ln\left(\frac{4x+1}{2x-7}\right)$

$-\frac{30}{(4x+1)(2x-7)}$

ii $(1+\ln x)^5$

$\frac{5}{x}(1+\ln x)^4$

jj $(\ln x - x)^9$

$9\left(\frac{1}{x}-1\right)(\ln x - x)^8$

kk $\frac{\ln x}{x}$

$\frac{1-\ln x}{x^2}$

ll $(2x+1)\ln x$

$\frac{2x\ln x + 2x + 1}{x}$

mm $x^3\ln(x+1)$

$3x^2\ln(x+1)+\frac{x^3}{x+1}$

nn $\sin 3x + \cos 8x$

$3\cos 3x - 8\sin 8x$

oo $\tan(\pi+x) + x^2$

$\sec^2(\pi+x) + 2x$

pp $x\tan x$

$x\sec^2 x + \tan x$

qq $\ln(\ln x)$

$\frac{1}{x\ln x}$

rr $\frac{\ln x}{x-2}$

$\frac{x-2-x\ln x}{x(x-2)^2}$

ss $\frac{e^{2x}}{\ln x}$

$\frac{e^{2x}(2x\ln x - 1)}{x(\ln x)^2}$

tt $e^x\ln x$

$e^x\left(\frac{1}{x}+\ln x\right)$

uu $5(\ln x)^2$

$\frac{10\ln x}{x}$

vv $\frac{e^{2x}}{\tan 7x}$

$\frac{e^{2x}(2\tan 7x - 7\sec^2 7x)}{\tan^2 7x}$

ww $\frac{e^{3x}}{x^2}$

$\frac{e^{3x}(3x-2)}{x^3}$

xx x^3e^{5x}

$x^2e^{5x}(5x+3)$

yy $\frac{e^{2x+1}}{2x+5}$

$\frac{4e^{2x+1}(x+2)}{(2x+5)^2}$

zz $\sin 2x\tan 3x$

$3\sin 2x\sec^2 3x + 2\tan 3x\cos 2x$

aaa $\frac{\sin x}{2x}$

$\frac{x\cos x - \sin x}{2x^2}$

bbb $\frac{3x+4}{\sin 5x}$

$\frac{3\sin 5x - 5(3x+4)\cos 5x}{\sin^2 5x}$

ccc $(2x + \tan 7x)^9$

$9(2 + 7\sec^2 7x)(2x + \tan 7x)^8$

ddd $\sin^2 x$

$\sin 2x$

eee $3\cos^3 5x$

$-45 \sin 5x \cos^2 5x$

fff $\log_2 x$

$\frac{1}{x \ln 2}$

ggg 2^x

$2^x \ln 2$

hhh $e^x + 3^x$

$e^x + 3^x \ln 3$

iii $e^x - \cos 2x$

$e^x + 2 \sin 2x$

jjj $\sin(1 - \ln x)$

$-\frac{1}{x}\cos(1 - \ln x)$

kkk $\sin(e^x + x)$

$(e^x + 1)\cos(e^x + x)$

lll $\sqrt{\sin 2x}$

$\frac{\cos 2x}{\sqrt{\sin 2x}}$

mmm $x + \log_3 x$

$1 + \frac{1}{x \ln 3}$

nnn $x^2 4^x$

$4^x(2x + x^2 \ln 4)$

ooo $\ln(\sin x)$

$\cot x$

ppp $e^{3x} \cos 2x$

$e^{3x}(3\cos 2x - 2\sin 2x)$

2 Differentiate with respect to x.

a $(\sin x - \cos x)^2$

$2(\sin^2 x - \cos^2 x)$

b $\sin^2 x + \cos^2 x$

0

c $\sin^2 x - \cos^2 x$

$4 \sin x \cos x$

d $\sin(\cos x)$

$-\sin x \cos(\cos x)$

e $\cos(\sin x)$

$-\cos x \sin(\sin x)$

f $\tan(\pi - x)$

$-\sec^2 x$

g $\sqrt{1 - \cos x}$

$\frac{\sin x}{2\sqrt{1 - \cos x}}$

h $\tan \sqrt{x}$

$\frac{1}{2\sqrt{x}} \sec^2 \sqrt{x}$

i $e^x \sin x$

$e^x(\sin x + \cos x)$

j $e^{2x} \cos\left(\frac{x}{2}\right)$

$e^{2x}\left(2 \cos\left(\frac{x}{2}\right) - \frac{1}{2} \sin\left(\frac{x}{2}\right)\right)$

k $e^{-x} \sin 3x$

$e^{-x}(3 \cos 3x - \sin 3x)$

l $e^x \cos 4x$

$e^x(\cos 4x - 4 \sin 4x)$

m $(\cos x + \sin x)e^{-x}$

$-2e^{-x}(\sin x)$

n $e^{\sin 2x}$

$2e^{\sin 2x} \cos 2x$

o $e^{\cos x}$

$-e^{\cos x} \sin x$

p $e^{\sin x + \cos x}$

$(\cos x - \sin x)e^{\sin x + \cos x}$

q $\log_e\left(\frac{x^3 - 1}{x}\right)$

$\frac{2x^3 + 1}{x(x^3 - 1)}$

r $\log_e(e^x(x + 2))$

$1 + \frac{1}{x + 2}$

s $x^3 \log_5 x$

$x^2\left(3 \log_5 x + \frac{1}{\ln 5}\right)$

t $\frac{2^x}{x}$

$\frac{2^x(x \ln 2 - 1)}{x^2}$

u $\frac{\log_a x}{x^2}$

$\frac{1 - 2x \ln a \log_a x}{x^3}$

v a^{-x}

$-a^{-x} \ln a$

w $a^x \log_a x$

$a^x \left(\ln a \log_a x + \frac{1}{x \ln a}\right)$

x $\frac{\log_a x}{a^x}$

$\frac{(1 - x \log_a x \times (\ln a)^2)}{x a^x \ln a}$

y $\sqrt{\log_a x}$

$\frac{1}{2x \ln a \sqrt{\log_a x}}$

z $\sqrt{x\, a^x}$

$\frac{\sqrt{(a^x)}(1 + x \ln a)}{2\sqrt{x}}$

aa $\left(x \log_2(x)\right)^2$

$= \frac{2x \log_2 x}{(\ln 2)((\ln 2)\log_2 x + 1)}$

bb $x\, e^{\sin x}$

$e^{\sin x} + x \cos x\, e^{\sin x}$

cc $e^x \log_e x$

$e^x \log_e x + \frac{e^x}{x}$

dd $e^{\cos(2x+1)}$

$-2 \sin(2x+1) e^{\cos(2x+1)}$

ee $1 + x + x^2 e^x$

$1 + 2xe^x + x^2 e^x$

ff $e^{\tan x}$

$e^{\tan x} \sec^2 x$

gg $e^{\sin 2x}$

$2e^{\sin 2x} \cos 2x$

hh $\sin(e^{2x})$

$2e^{2x} \cos(e^{2x})$

ii $\log_e(\cos 4x)$

$-4 \tan 4x$

jj $\log_e(\sin x)$

$\cot x$

3 If $y = e^x \sin x$, find y' and y'', and show that $y'' - 2y' + 2y = 0$.

$$y' = e^x \sin x + e^x \cos x$$
$$= e^x(\sin x + \cos x)$$
$$y'' = e^x(\sin x + \cos x) + e^x(\cos x - \sin x)$$
$$= e^x(2\cos x)$$

$$y'' - 2y' + 2y = 2e^x \cos x - 2e^x(\sin x + \cos x) + 2e^x \sin x$$
$$= 2e^x \cos x - 2e^x \sin x - 2e^x \cos x + 2e^x \sin x$$
$$= 0$$

4 If $y = e^{-x} \cos x$, find y' and y'', and show that $y'' + 2y' + 2y = 0$.

$$y' = -e^{-x} \cos x + e^{-x}(-\sin x) = -e^{-x}(\cos x + \sin x)$$
$$y'' = e^{-x}(\cos x + \sin x) + (-e^{-x})(-\sin x + \cos x)$$
$$= e^{-x}(\cos x + \sin x) - e^{-x}(\cos x - \sin x)$$
$$= e^{-x}(2\sin x)$$

$$y'' + 2y' + 2y = 2e^{-x} \sin x + 2(-e^{-x})(\cos x + \sin x) + 2e^{-x} \cos x$$
$$= 2e^{-x} \sin x - 2e^{-x} \cos x - 2e^{-x} \sin x + 2e^{-x} \cos x$$
$$= 0$$

5 If $f(x) = \log_e\left(\frac{1 + \sin x}{\cos x}\right)$, show that $f'(x) = \sec x$.

$$f(x) = \ln(1 + \sin x) - \ln(\cos x)$$
$$f'(x) = \frac{\cos x}{1 + \sin x} - \frac{-\sin x}{\cos x}$$
$$= \frac{\cos x}{1 + \sin x} + \frac{\sin x}{\cos x}$$
$$= \frac{\cos^2 x + \sin x(1 + \sin x)}{\cos x(1 + \sin x)}$$
$$= \frac{\cos^2 x + \sin x + \sin^2 x}{\cos x(1 + \sin x)}$$
$$= \frac{1 + \sin x}{\cos x(1 + \sin x)}$$
$$= \frac{1}{\cos x}$$
$$= \sec x$$

6 If $y = \ln(\tan 2x)$, show that $\frac{dy}{dx} = 2\sec 2x \operatorname{cosec} 2x$.

$$\frac{dy}{dx} = \left(\frac{1}{\tan 2x}\right) \times \sec^2 2x \times 2$$
$$= \frac{2\sec^2 2x}{\tan 2x}$$
$$= \frac{2\sec^2 2x}{\left(\frac{\sin 2x}{\cos 2x}\right)}$$
$$= \frac{2\sec^2 2x \times \cos 2x}{\sin 2x}$$
$$= \frac{2\sec 2x}{\sin 2x}$$
$$= 2\sec 2x \operatorname{cosec} 2x$$

7 The point $P\left(\frac{\pi}{6}, \frac{1}{2}\right)$ lies on the curve $y = 2\sin x - \cos 2x$.

a Show that the tangent at P has equation $2\sqrt{3}x - y = \frac{\pi\sqrt{3}}{3} - \frac{1}{2}$.

$\frac{dy}{dx} = 2\cos x + 2\sin 2x$

At $x = \frac{\pi}{6}$:

$\frac{dy}{dx} = 2\cos\left(\frac{\pi}{6}\right) + 2\sin\left(\frac{\pi}{3}\right) = 2\left(\frac{\sqrt{3}}{2}\right) + 2\left(\frac{\sqrt{3}}{2}\right) = 2\sqrt{3}$

Tangent: $y - \frac{1}{2} = 2\sqrt{3}\left(x - \frac{\pi}{6}\right)$

$y = 2\sqrt{3}x - \frac{\pi\sqrt{3}}{3} + \frac{1}{2}$

$2\sqrt{3}x - y = \frac{\pi\sqrt{3}}{3} - \frac{1}{2} = \frac{\pi\sqrt{3}}{3} - \frac{1}{2}$

b Show that the normal at P has equation $x + 2\sqrt{3}y = \frac{\pi}{6} + \sqrt{3}$.

Normal has gradient $-\frac{1}{2\sqrt{3}}$

Normal: $y - \frac{1}{2} = \left(-\frac{1}{2\sqrt{3}}\right)\left(x - \frac{\pi}{6}\right)$

$2\sqrt{3}y - \sqrt{3} = -x + \frac{\pi}{6}$

$x + 2\sqrt{3}y = \frac{\pi}{6} + \sqrt{3}$

8

a Differentiate $y = e^{\sin x}$.

$y' = \cos x\, e^{\sin x}$

b Hence find, in the domain $0 \leq x \leq 2\pi$, the x-coordinates of the points on the curve $y = e^{\sin x}$ where the tangent is horizontal.

$\cos x\, e^{\sin x} = 0$

Since $e^{\sin x} > 0$, need $\cos x = 0$

In $[0, 2\pi]$: $x = \frac{\pi}{2}, \frac{3\pi}{2}$

9

a Differentiate $y = e^{\cos x}$.

$y' = e^{\cos x} \times (-\sin x) = -\sin x\, e^{\cos x}$

b Hence find, in the domain $0 \leq x \leq 2\pi$, the x-coordinates of the points on the curve $y = e^{\cos x}$ where the tangent is horizontal.

$-\sin x\, e^{\cos x} = 0$

Since $e^{\cos x} > 0$, need $\sin x = 0$

In $[0, 2\pi]$: $x = 0, \pi, 2\pi$

10 If $y = \ln\left(\frac{\sqrt{2}-\cos x}{\sqrt{2}+\cos x}\right)$, show that $\frac{dy}{dx} = \frac{2\sqrt{2}\sin x}{1+\sin^2 x}$.

$$y = \ln(\sqrt{2} - \cos x) - \ln(\sqrt{2} + \cos x)$$

$$\frac{dy}{dx} = \frac{\sin x}{\sqrt{2} - \cos x} - \frac{-\sin x}{\sqrt{2} + \cos x}$$

$$= \frac{\sin x}{\sqrt{2} - \cos x} + \frac{\sin x}{\sqrt{2} + \cos x}$$

$$= \frac{\sin x(\sqrt{2} + \cos x) + \sin x(\sqrt{2} - \cos x)}{(\sqrt{2} - \cos x)(\sqrt{2} + \cos x)}$$

$$= \frac{\sin x(2\sqrt{2})}{2 - \cos^2 x}$$

$$= \frac{2\sqrt{2}\sin x}{2 - \cos^2 x}$$

$$= \frac{2\sqrt{2}\sin x}{1 + 1 - \cos^2 x}$$

$$= \frac{2\sqrt{2}\sin x}{1 + \sin^2 x}$$

11

a Find the equation of the tangent to the curve $y = 10^x$ at the point (1, 10).

$$\frac{dy}{dx} = 10^x \ln 10$$

$$At\ (1, 10): m = 10 \ln 10$$

$$\text{Tangent: } y - 10 = (10 \ln 10)(x - 1)$$

$$(10 \ln 10)x - y + 10(1 - \ln 10) = 0$$

b Find the equation of the normal to the curve $y = 5^x$ at the point (2, 25).

$$\frac{dy}{dx} = 5^x \ln 5$$

$$\text{At } (2, 25): m = 25 \ln 5$$

$$\text{Normal gradient} = -\frac{1}{25 \ln 5}$$

$$\text{Normal: } y - 25 = \left(-\frac{1}{25 \ln 5}\right)(x - 2)$$

$$25y \ln 5 - 625 \ln 5 = -x + 2$$

$$x + 25y \ln 5 - 625 \ln 5 - 2 = 0$$

c At what x value are the tangents to $y = 10^x$ and $y = 5^x$ parallel? Answer to 3 s.f.

Tangents parallel when:

$$10^x \ln 10 = 5^x \ln 5$$

$$\left(\frac{10}{5}\right)^x = \frac{\ln 5}{\ln 10}$$

$$2^x = \frac{\ln 5}{\ln 10}$$

$$2^x = \log_{10} 5$$

$$x = \log_2(\log_{10} 5))$$

$$x \approx -0.517$$

https://www.desmos.com/calculator/ktu3r3zaby

12

a Show that $y = \sin^2 x$ has derivative $y' = 2 \sin x \cos x$.

$y' = 2 \sin x \times \cos x = 2 \sin x \cos x$

b Find the gradients of the tangent and normal to $y = \sin^2 x$ at the point where $x = \frac{\pi}{4}$.

At $x = \frac{\pi}{4}$:

$y' = 2 \sin\left(\frac{\pi}{4}\right) \cos\left(\frac{\pi}{4}\right) = 2\left(\frac{\sqrt{2}}{2}\right)\left(\frac{\sqrt{2}}{2}\right) = 1$

Gradient of tangent $= 1$

Gradient of normal $= -1$

c Find the equations of the tangent and normal to $y = \sin^2 x$ at the point where $x = \frac{\pi}{4}$.

At $x = \frac{\pi}{4}$: $y = \sin^2\left(\frac{\pi}{4}\right) = \frac{1}{2}$

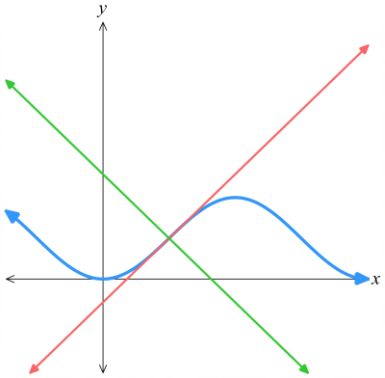
Tangent: $y - \frac{1}{2} = 1\left(x - \frac{\pi}{4}\right)$

$x - y = \frac{\pi}{4} - \frac{1}{2}$

Normal: $y - \frac{1}{2} = -1\left(x - \frac{\pi}{4}\right)$

$x + y = \frac{\pi}{4} + \frac{1}{2}$

d Suppose that the tangent meets the x-axis at P, the normal meets the y-axis at Q and O is the origin. Show that $\triangle OPQ$ has area $\frac{(\pi^2 - 4)}{32}$ units².



Tangent $x - y = \frac{\pi}{4} - \frac{1}{2}$ meets x axis $(y = 0)$:

$x = \frac{\pi}{4} - \frac{1}{2}$, so $P = \left(\frac{\pi}{4} - \frac{1}{2}, 0\right)$

Normal $x + y = \frac{\pi}{4} + \frac{1}{2}$ meets y axis $(x = 0)$:

$y = \frac{\pi}{4} + \frac{1}{2}$, so $Q = \left(0, \frac{\pi}{4} + \frac{1}{2}\right)$

$\text{Area} = \left(\frac{1}{2}\right)|OP||OQ| = \left(\frac{1}{2}\right)\left(\frac{\pi}{4} - \frac{1}{2}\right)\left(\frac{\pi}{4} + \frac{1}{2}\right)$

$= \left(\frac{1}{2}\right)\left(\left(\frac{\pi}{4}\right)^2 - \left(\frac{1}{2}\right)^2\right)$

$= \left(\frac{1}{2}\right)\left(\frac{\pi^2}{16} - \frac{1}{4}\right)$

$= \left(\frac{1}{32}\right)(\pi^2 - 4)$

$= \frac{(\pi^2 - 4)}{32}$

Exam Questions

1 HSC Advanced Sample Question Band 4

If $f(x) = \log_2(x^{2x})$, which expression is equal to $f'(x)$?

A. $\dfrac{2}{x^{2x}\ln 2}$

B. $\dfrac{2}{\ln 2} + 2\log_2 x$

C. $\log_2 x + \dfrac{2}{\ln 2}$

D. $\dfrac{2}{\ln 2} \times \log_2\left(x^{2x-1}\right)$

$f(x) = 2x\log_2 x = \dfrac{2x\ln x}{\ln 2}$

$= \dfrac{2}{\ln 2}\, x\ln x$

$f'(x) = \dfrac{2}{\ln 2}(\ln x + 1)$

$= \dfrac{2\ln x}{\ln 2} + \dfrac{2}{\ln 2}$

$= 2\log^2 x + \dfrac{2}{\ln 2}$

B

2 HSC Advanced Sample Question Band 4

If $f(x) = e^{g(x^2)}$, where g is a differentiable function, find $f'(x)$.

A. $2xe^{g(x^2)}$

B. $2xg\left(x^2\right)e^{g(x^2)}$

C. $2xg'\left(x^2\right)e^{g(x^2)}$

D. $2xg'(2x)e^{g(x^2)}$

$f'(x) = \dfrac{d[g(x^2)]}{dx} \cdot e^{g(x^2)}$

$= 2x \cdot g'(x^2) \cdot e^{g(x^2)}$

C

3 HSC Advanced Sample Question Band 5

Given $f(x) = \log_{10} x^x$, which expression is equal to $f'(x)$?

A. $\log_e 10 + \log_e x$

B. $\dfrac{\log_e 10 + 1}{\log_e 10}$

C. $\dfrac{1}{\log_e 10} + \log_x 10$

D. $\dfrac{1}{\log_e x} + \log_{10} x$

$f(x) = x\log_{10} x$

$f'(x) = x \cdot \dfrac{1}{x\ln 10} + 1 \cdot \log_{10} x$

$= \dfrac{1}{\ln 10} + \log_{10} x$

$= \dfrac{1}{\ln 10} + \dfrac{\ln x}{\ln 10}$

$= \dfrac{\ln x + 1}{\ln 10}$

B

4 HSC Advanced Sample Question Band 4

Differentiate π^{2x}

$y = \pi^{2x}$

$\dfrac{dy}{dx} = \log_e(\pi) \cdot 2 \cdot \pi^{2x}$

$= 2\log_e(\pi) \cdot \pi^{2x}$

5 HSC Advanced Sample Question Band 4

Differentiate $5^{x^2} \cdot 5x$

$$\frac{dy}{dx} = \ln 5 \cdot 2x \cdot 5^{x^2} \cdot 5x + 5^{x^2} \cdot 5$$
$$= 5^{x^2}(\ln 5 \cdot 10x^2 + 5)$$
$$= 5^{x^2+1}(\ln 5 \cdot 2x^2 + 1)$$

6 HSC Advanced Sample Question Band 4

Differentiate with respect to x: $\log_e x^x$

$$\frac{dy}{dx} = x \cdot \frac{1}{x} + \log_e x$$
$$= 1 + \log_e x$$

7 2024 HSC Advanced Band 4

Let $f(x) = \log_e(x^3 - 3x + 2)$. Find $f'(3)$

$$f'(x) = \frac{3x^2 - 3}{x^3 - 3x + 2}$$
$$f'(3) = \frac{6}{5}$$

8 2013 HSC Advanced Band 4

What is the derivative of $\frac{x}{\cos x}$?

A. $\frac{\cos x + x\sin x}{\cos^2 x}$

B. $\frac{\cos x - x\sin x}{\cos^2 x}$

C. $\frac{x\sin x - \cos x}{\cos^2 x}$

D. $\frac{-x\sin x - \cos x}{\cos^2 x}$

$$\frac{dy}{dx} = \frac{vu' - uv'}{v^2}$$
$$= \frac{\cos x \cdot 1 - x(-\sin x)}{(\cos x)^2}$$
$$= \frac{\cos x + x \sin x}{\cos^2 x}$$

A

9 HSC Advanced Sample Question Band 4

Consider the function $f(\theta) = sin^3(2\theta)$.

If $f'(\theta) = 6\cos(2\theta) - 6\cos^n(2\theta)$, find the value of n.

$$f(\theta) = sin^3(2\theta) = \left(sin(2\theta)\right)^3$$

Using chain rule:

$$f'(\theta) = 3 \times 2cos(2\theta) \times sin^2(2\theta)$$
$$= 6cos(2\theta)\left(1 - cos^2(2\theta)\right)$$
$$= 6cos(2\theta) - 6cos^3(2\theta)$$
$$\therefore n = 3$$

10 2007 HSC Advanced Band 4

The point $P(\pi, 0)$ lies on the curve $y = x \sin x$.
Find the equation of the tangent to the curve at P.

$\frac{dy}{dx} = x\cos x + \sin x$

When $x = \pi$

$\frac{dy}{dx} = \pi \times \cos\pi + \sin\pi$

$= \pi(-1) + 0$

$= -\pi$

Equation of line, $m = -\pi$, through $P(\pi, 0)$:

$y - y_1 = m(x - x_1)$

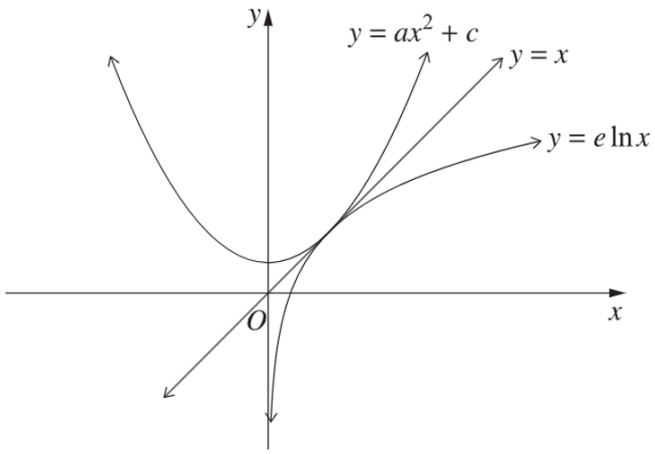
$y - 0 = -\pi(x - \pi)$

$\therefore y = -\pi x + \pi^2$

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The graphs of $y = e\ln x$ and $y = ax^2 + c$ are shown.
The line $y = x$ is a tangent to both graphs at their point of intersection.

Find the values of a and c.



$y = e\ln x \ldots(1)$ $y = ax^2 + c \ldots(2)$

$y' = \frac{e}{x}$ $y' = 2ax$

At point of tangency, gradient = 1.

Find x such that:

$\frac{e}{x} = 1 \Rightarrow x = e$

At $x = e, y' = 2ax = 1$:

$2ae = 1 \Rightarrow a = \frac{1}{2e}$

Find point of tangency using (1):

$y = e\ln e = e$

Point of tangency at (e, e).

Since (e, e) lies on (2):

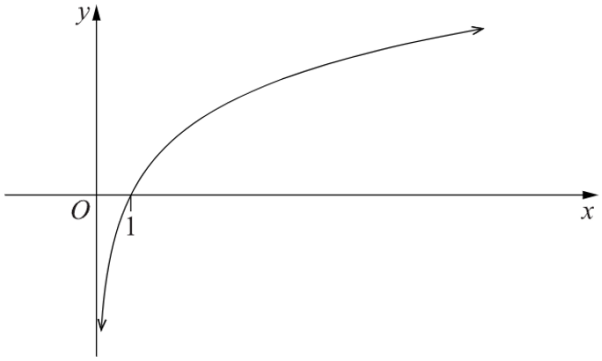
$e = \left(\frac{1}{2e}\right) \times e^2 + c$

$c = \frac{e}{2}$

$\therefore a = \frac{1}{2e}, c = \frac{e}{2}$

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The diagram shows the graph of y = c ln x, c > 0.



a Show that the equation of the tangent to y = c ln x at x = p, where p > 0, is

$$y = \frac{cx}{p} - c + c \ln p$$

b Find the value of *c* such that the tangent from part **a** has a gradient of 1 and passes through the origin.

$y = c \ln x$

$\frac{dy}{dx} = \frac{c}{x}$

At $x = p$,

$m = \frac{c}{p}$

Tangent passes through $(p, c \ln p)$

$\therefore$ Equation of tangent

$y - c \ln p = \left(\frac{c}{p}\right)(x - p)$

$y = \left(\frac{c}{p}\right)x - c + c \ln p$

If m = 1,

$\frac{c}{p} = 1$

$c = p$

If tangent passes through $(0,0)$

$0 = -c + c \ln c$

$0 = c(\ln c - 1)$

$\ln c = 1 \ (c > 0)$

$\therefore c = e$

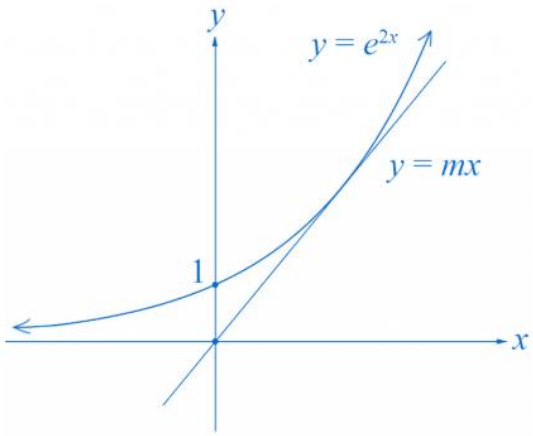
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The line $y = mx$ is a tangent to the curve $y = e^{2x}$ at a point P.

a Sketch the line and the curve on one diagram.

b Find the coordinates of P.

c Find the value of m.



$y = e^{2x}$

$\frac{dy}{dx} = 2e^{2x}$

Gradient of $y = mx$ is m

Gradients equal when

$2e^{2x} = m$

$e^{2x} = \frac{m}{2}$

$\ln e^{2x} = \ln\left(\frac{m}{2}\right)$

$2x = \ln\left(\frac{m}{2}\right)$

$x = \left(\frac{1}{2}\right)\ln\left(\frac{m}{2}\right)$

When $x = \left(\frac{1}{2}\right)\ln\left(\frac{m}{2}\right)$

$y = e^{2\times\left(\frac{1}{2}\right)ln\left(\frac{m}{2}\right)}$

$= e^{\ln\left(\frac{m}{2}\right)}$

$= \frac{m}{2}$

$\therefore P\left(\frac{1}{2}ln\left(\frac{m}{2}\right), \frac{m}{2}\right)$

$y = mx$ passes through $(0,0)$ and $\left(\left(\frac{1}{2}\right)ln\left(\frac{m}{2}\right), \frac{m}{2}\right)$

Using gradient formula:

$\frac{\frac{m}{2} - 0}{\left(\frac{1}{2}\right)ln\left(\frac{m}{2}\right) - 0} = m$

$\frac{\left(\frac{m}{2}\right)}{\frac{1}{2}\ln\left(\frac{m}{2}\right)} = m$

$\frac{m}{\ln\left(\frac{m}{2}\right)} = m$

$\ln\left(\frac{m}{2}\right) = 1$

$\frac{m}{2} = e^1$

$m = 2e$